

Plyometric Exercise and Aerobic Exercise

Many sports—such as basketball and soccer—have both an anaerobic (i.e., power) and an aerobic component. Therefore, multiple types of training must be combined to best prepare athletes for these types of sports. Because aerobic exercise may have an interfering effect on power production (17), it is advisable to perform plyometric exercise before aerobic endurance training. The program design variables do not change and should complement each other to most effectively prepare these athletes for competition.

Safety Considerations

Plyometric exercise is not inherently dangerous; however, as with all modes of exercise, the risk of injury exists. Injuries can occur simply due to an accident, but they more typically occur when proper training procedures are violated and may be the result of an insufficient strength and conditioning base, inadequate warm-up, improper progression of lead-up drills, inappropriate volume or intensity for the phase of training, poor shoes or surface, or a simple lack of skill. The following sections identify and address these and other risk factors. Knowledge of risk factors can improve the safety of athletes performing plyometric exercise.

Pretraining Evaluation of the Athlete

To reduce the risk of injury and facilitate the performance of plyometric exercises, the athlete must understand proper plyometric technique and possess a sufficient base of strength, speed, and balance. In addition, the athlete must be sufficiently mature, both physically and psychologically, to participate in a plyometric training program. The following evaluative items can help determine whether an athlete meets these conditions.

Technique

Before adding any drill to an athlete's plyometric program, the strength and conditioning professional must first demonstrate proper technique to the athlete and then coach and cue the athlete on proper technique to maximize the drill's effectiveness and minimize the risk of injury. For lower body plyometrics, proper landing technique is essential. If an athlete's center of gravity is offset from the base of support, performance is hindered, and injury may occur. The shoulders should be over the knees and the knees should be over the toes during the landing, which the athlete accomplishes through ankle dorsiflexion and knee and hip flexion ([figure 19.3](#)). In addition, when viewing the frontal plane and motion of the athlete

performing lower body plyometrics, it is essential that the knees are positioned over the toes. Inward movement of the knees—also termed **dynamic valgus**—is a significant risk factor for knee injuries of all types, including patellofemoral pain and tears or ruptures of the anterior cruciate ligament (ACL) (45).

In addition to instructing an athlete on a proper body position while landing, the strength and conditioning professional must explain, coach, and cue the athlete to land with an **active flat foot**. In preparation for ground contact, the athlete's ankles should be dorsiflexed (so that the toes are pulled back toward the knees). This means that the bottom of the foot is parallel to the ground as the athlete comes into landing. Immediately before landing, the athlete should be taught to aggressively plantarflex the ankle to "attack the ground with the foot." This achieves two outcomes. First, the plantarflexion increases the tension in the gastrocnemius–soleus complex at ground contact, which increases the efficacy of the SSC and enhances the output power in the resulting jump. Second, the last-second adjustment in the angle of the foot, along with the increased muscle tension, means that the heels stay off the ground during the land–takeoff action. This enhances both anterior and vertical propulsion and is often coached as "the credit card rule," meaning that enough of the foot should be in contact with the ground to achieve high impulses, but the strength and conditioning professional should be able to slide a credit card under the athlete's heels (12a).

Strength

Consideration of the athlete's level of strength is necessary before he or she performs plyometrics. Strength is essential for the maintenance of posture through the plyometric action and the ability to manage and appropriately direct ground reaction forces through effective ground contact to optimize performance benefits and reduce potential for injury ([figure 19.4](#)). Without these outcomes, safe and effective techniques cannot be maintained as the intensity

of plyometric exercise is increased. Previous recommendations for lower body plyometrics stated that the athlete's 1RM squat should be at least 1.5 times his or her body weight (17, 24, 36, 50, 63). However, many plyometric activities can be safely taught to athletes without such formalized levels of strength ([figure 19.5](#)). It is our recommendation that plyometrics be included for all athletes whose sports require running, landing, jumping, or cutting. As a prerequisite, teaching proper alignment and movement mechanics through the use of plyometric exercise has been shown repeatedly to decrease the athlete's risk of injury during practices and games (50). Various authors have theorized that basic in-place jumps such as repeated pogo jumps and countermovement jumps require relatively low levels of strength; repeated hops and jumps may require a basic strength level equivalent to a back squat of $1 \times$ body weight, bounding and depth jumps using a box 12 inches (30 cm) tall or shorter may have a back squat strength equivalent of $1.5 \times$ body weight, and depth or box jumps with a box higher than 12 inches (30 cm) may require a back squat strength equivalent of $2 \times$ body weight (58).

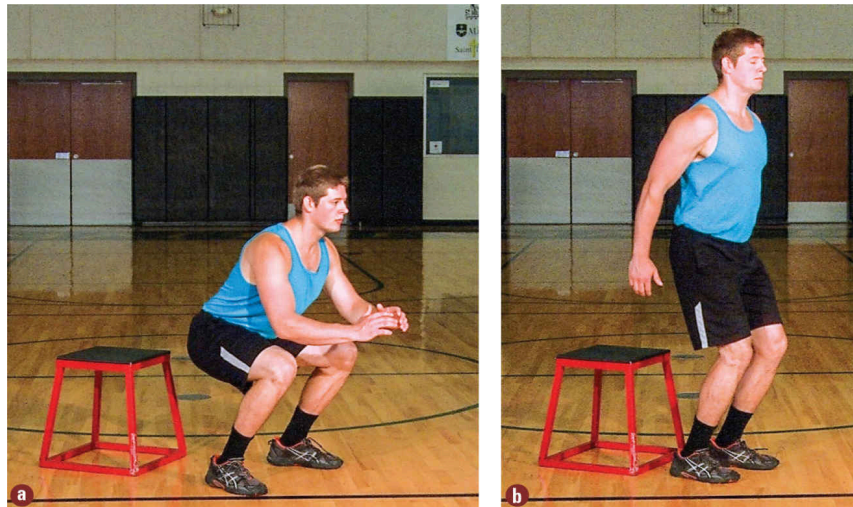


FIGURE 19.3 Proper plyometric landing technique. (a) When learning to land and stop, the shoulders are in line with the knees, which helps place the center of gravity over the body's base of support. The knees are over the toes; excessive inward (valgus) movement increases the risk of lower extremity injury; weight is evenly distributed through a flat foot to allow a balanced deceleration position. Learning to land and stop, decelerating body mass completely, is a precursor to learning to land and rebound (i.e., reaccelerate body mass). (b) Proper landing technique to facilitate reactionary rebound actions; the shoulders are in line with the knees, the ankles are stiff, and athlete contacts the ground with the midfoot and forefoot to enable a rebound acceleration to occur. Note the alignment of the hips, knees and toes, with the feet hip-width apart to optimize vertical or horizontal force (depending on the subsequent movement).

Back squat equivalent strength is a descriptive term used to infer the necessary strength capacity required to tolerate the impact stress of the exercise so that it does not acutely overreach the athlete and become injurious. The back squat is illustrative due to the nature of its axial load on the total body, yet it is recognized that not every athlete measures maximal squat strength. It is important to note that it is not simply the magnitude of strength that is important but the athletic quality to be able to apply the force required within the given ground contact time. Therefore, a focus on rate of force development as well as magnitude of force development is an important consideration within plyometric prescription.

Balance

Balance is a lower body plyometric prerequisite that should not be overlooked. **Balance** is the maintenance of a position without movement for a given period of time. Many lower body plyometric drills require the athlete to move in nontraditional patterns or on a single leg. These types of drills necessitate a solid, stable base of support on which the athlete can safely and correctly perform the exercises. Three balance tests are provided in [table 19.9](#), listed in order of difficulty; each test position must be held for 30 seconds (62). For example, an athlete beginning plyometric training for the first time would be required to stand on one leg for 30 seconds without falling. An experienced athlete beginning an advanced plyometric training program must maintain a single-leg half squat for 30 seconds without falling. Additionally, the surface on which the balance testing is performed should be the same as that used in the plyometric drills.

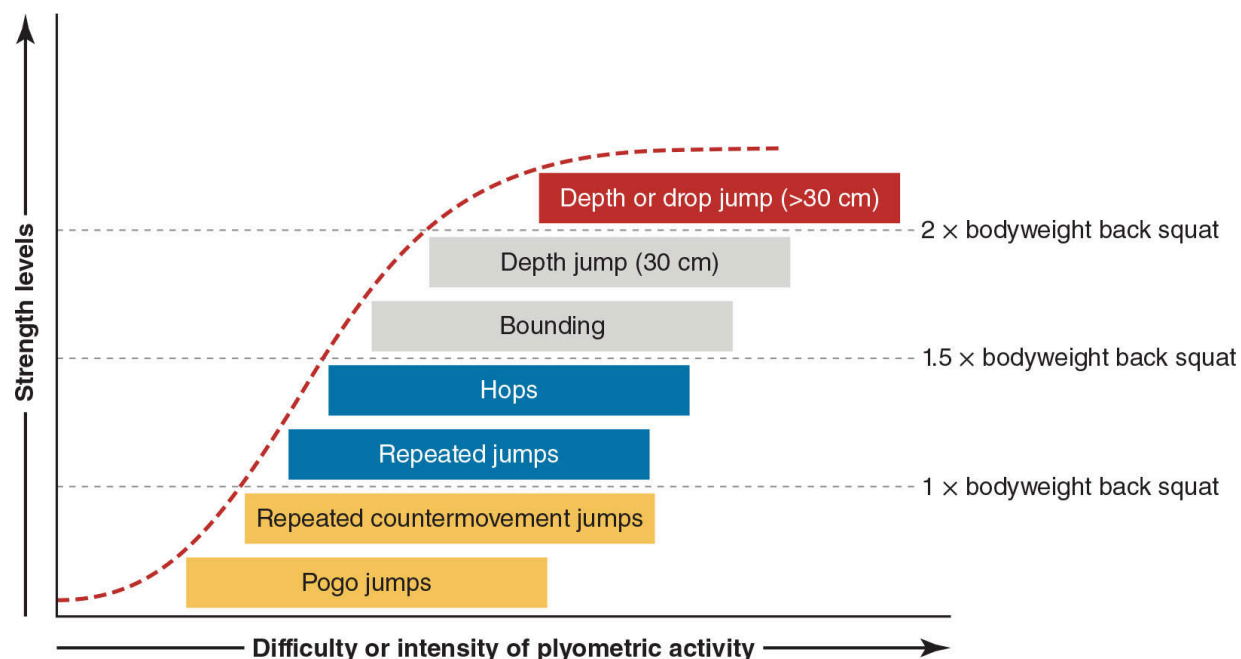


FIGURE 19.5 Theoretical relationship between strength and the optimization of plyometric exercise performance.

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TABLE 19.9 Balance Tests

Test*	Variations
Standing	Double leg
	Single leg
Quarter squat	Double leg
	Single leg
Half squat	Double leg
	Single leg

*A successful test occurs when the position can be held for 30 seconds without falling.

Physical Characteristics

Athletes who weigh more than 220 pounds (100 kg) may be at an increased risk for injury when performing plyometric exercises (50,

63). Greater weight (mass) increases the compressive force on joints during the exercises, thereby predisposing ankle, knee, and hip–pelvic complex joints to injury. Therefore, athletes weighing over 220 pounds (100 kg) should approach high-volume and high-intensity plyometric exercises and depth jumps from heights greater than 18 inches (46 cm) with caution (50, 63). As with other forms of exercise, an athlete's joint structure and previous injuries must also be examined before he or she begins a plyometric training program. Previous injuries or abnormalities of the spine, lower extremities, or upper extremities may increase an athlete's risk of injury during plyometric exercise. Specifically, athletes with a history of muscle strains, pathological joint laxity, or spinal dysfunction—including vertebral disk dysfunction or compression—should use caution when beginning a plyometric training program (28, 29, 36, 55). Ultimately, because the body composition, strength, and technical competency can differ between athletes who weigh the same, professional judgment always should be applied when considering plyometric prescription.

Equipment and Facilities

In addition to athletes' fitness and health, the area and equipment used for plyometric drills may significantly affect their safety.

Landing Surface

To prevent injuries, the landing surface used for lower body plyometrics must possess adequate shock-absorbing properties. A grass field, suspended floor, or rubber mat is a good surface choice (36). Surfaces such as concrete, tile, and hardwood are not recommended because they lack effective shock-absorbing properties (36). Excessively thick exercise mats (6 inches [15 cm] or thicker) may extend the amortization phase and thus not allow efficient use of the stretch reflex. Mini-trampolines are commonly used for beginning plyometric and balance training in rehabilitation (32). While these devices may provide a necessary introduction to

plyometrics, especially for those recovering from musculoskeletal injury, mini-trampolines, like thick exercise mats, are not effective for plyometric training of uninjured athletes because the amortization phase is extended while the athlete is in contact with the elastic surface.

Training Area

The amount of space needed depends on the drill. Most bounding and running drills require at least 30 yards (or meters) of straightaway, though some drills may require a straightaway of 100 yards (or meters). For most standing, box, and depth jumps, only a minimal surface area is needed, but the ceiling height must be at least 10 feet (3 m) to be adequate.

Equipment

Boxes used for box jumps and depth jumps must be sturdy and should have a nonslip top. Boxes should range in height from 6 to 42 inches (15-107 cm) and should have landing surfaces of at least 18 by 24 inches (46 × 61 cm) (19). The box should be constructed of sturdy wood (e.g., 3/4-inch [1.9 cm] plywood) or heavy-gauge metal that is covered in a protective covering to reduce the risk of injury if the athlete contacts the box. High density foam within a cubed framework with solid landing surfaces is also appropriate. To further reduce injury risk, there are several ways of making the landing surface nonslip such as adding nonslip treads, mixing sand into the paint used to cover the box, or affixing rubberized flooring to the top (19).

Proper Footwear

Athletes must use footwear with good ankle and foot support; good lateral stability; and a wide, nonslip sole (50). Ankle injury risk may be increased in shoes with a narrow sole and poor mid-foot support, especially with lateral movements. Shoes with insufficient foot support may lead to arch or lower leg injuries or both, while footwear without enough cushioning might lead to knee and hip

joint damage. Shoes with excessive cushioning may also interfere with the amortization phase process, and a shoe that has increased inbuilt pronation (such as a weightlifting shoe) may interfere with ground contact mechanics and therefore the safety and effectiveness of the exercise.

Supervision

In addition to the safety considerations already outlined, close monitoring of athletes is necessary to ensure proper technique. Plyometric exercise is not intrinsically dangerous when performed correctly, but as with other forms of training, poor technique may unnecessarily predispose an athlete to injury.

Depth Jumping

There is a limit to the maximal height at which a depth jump can be effectively and safely performed. A height of 48 inches (122 cm) would provide a significant overload on the muscles, but the resistance may be too great for many athletes to overcome while maintaining correct technique (47). Jumping from such a height increases the possibility of injury; furthermore, the amount of force to be overcome is so great that the amortization phase is extended and thus the purpose of the exercise is defeated. The recommended height for depth jumps ranges from 16 to 42 inches (41-107 cm), with 30 to 32 inches (76-81cm) being the norm (4, 21, 30, 43, 44, 48). The depth jump box height for athletes who weigh over 220 pounds (100 kg) should be 18 inches (46 cm) or less.

Conclusion

The major goal of plyometric training is to rapidly apply force to provide an overload to the agonist muscles. Although it has been repeatedly shown that plyometric exercise increases muscular power for athletes in a formal training program (34, 53, 65, 67), research has yet to determine whether mechanical or neurophysiological adaptations account for the improvement. Plyometrics should be considered not an end in itself but part of an appropriately periodized program that includes strength, speed, endurance, and mobility training. Plyometrics requires an appropriate warm-up and should be preplanned to allow adequate recovery. After the athlete has begun a proper strength and conditioning program, plyometric training may be used to further develop impulses and positive transfer of training benefits.

PLYOMETRIC DRILLS

Lower Body Plyometric Drills

JUMPS IN PLACE

- 1 Pogo Jump [582](#)
- 2 Ankle Hop [582](#)
- 3 Squat Jump [583](#)
- 4 Jump and Reach [584](#)
- 5 Tuck Jump [585](#)
- 6 Split Squat Jump [586](#)
- 7 Cycled Split Squat Jump [587](#)
- 8 Tuck Hop [588](#)
- 9 Pike Jump [589](#)

STANDING JUMPS

- 10** Vertical Jump [590](#)
- 11** Vertical Hop [591](#)
- 12** Jump Over Barrier [591](#)
- 13** Standing Long Jump [592](#)

MULTIPLE HOPS AND JUMPS

- 14** Multiple Horizontal Low Hurdle Jump [593](#)
- 15** Double-Leg Zigzag Jump [594](#)
- 16** Linear Hop [595](#)
- 17** Multiple Barrier Hop [596](#)
- 18** Lateral Barrier Jump [597](#)
- 19** Four-Hurdle Hop [598](#)
- 20** Progressive Lateral Jump [599](#)

BOUNDS

- 21** Skip [600](#)
- 22** Power Skip [601](#)
- 23** Lateral Skip [602](#)
- 24** Linear Bound [603](#)
- 25** Double-Arm Linear Bound [604](#)

BOX DRILLS

- 26** Drop Step to Lateral Bound and Return [605](#)
- 27** Single-Leg Push-Off [606](#)
- 28** Alternating-Leg Push-Off [607](#)
- 29** Lateral Push-Off [608](#)
- 30** Rear Foot–Elevated Multidirectional Hop [609](#)
- 31** Side-to-Side Push-Off [610](#)
- 32** Box Jump [611](#)
- 33** Box Hop [612](#)
- 34** Lateral Box Jump [613](#)

35 Drop Freeze [614](#)

DEPTH JUMPS

36 Depth Jump [615](#)

37 Depth Jump to Box Jump [616](#)

38 Depth Jump Into Directional Sprint [617](#)

39 Depth Jump to Long Jump [618](#)

40 Depth Jump to 180-Degree Turn [619](#)

41 Depth Hop [620](#)

Upper Body Plyometric Drills

THROWS

42 Chest Pass [621](#)

43 Two-Hand Overhead Throw [622](#)

44 Rotational Throw [623](#)

45 Single-Arm Throw [624](#)

46 Power Drop [625](#)

47 Half-Kneeling Overhead Medicine Ball Throw for Distance
[626](#)

48 Single-Arm Reverse Throw [626](#)

49 Viking Throw for Height [628](#)

50 Medicine Ball Slam from Single-Leg Balance [629](#)

PLYOMETRIC PUSH-UPS

51 Depth Push-Up [630](#)

52 Lateral Travel Push-Up [631](#)

TRUNK PLYOMETRICS

53 45-Degree Sit-Up [632](#)

54 Seated Rotational Medicine Ball Throw [633](#)

55 Leg Pushdown [634](#)

LOWER BODY PLYOMETRIC DRILLS

Jumps in Place

1 POGO JUMP

Intensity level: Low

Direction: Vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: None or double arm

Preparatory movement: Begin with a slight countermovement.

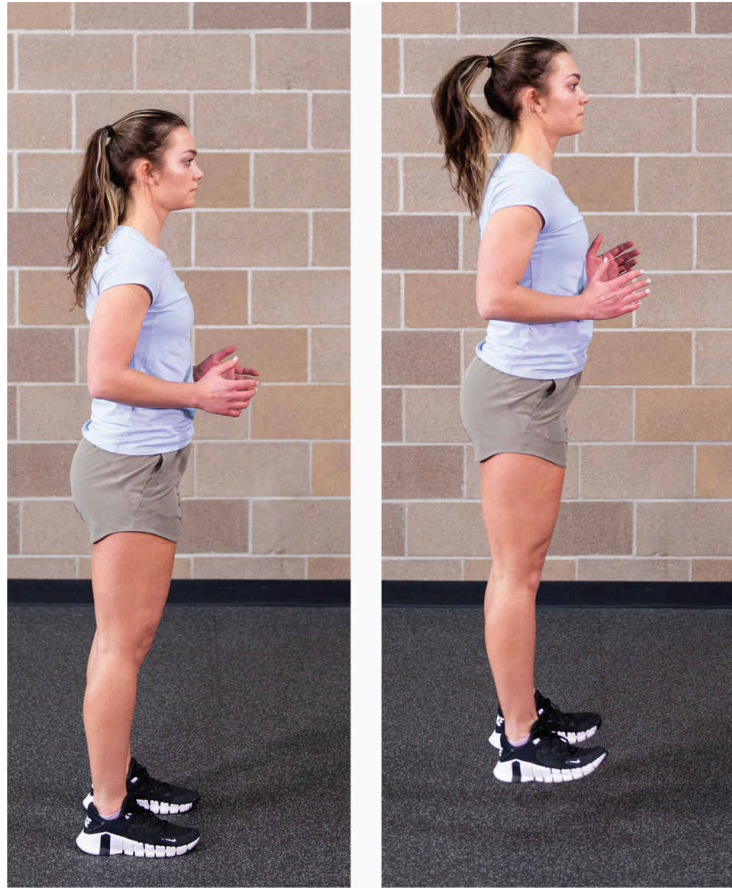
Upward movement: Jump up, with primary motion at the ankle joint.

Downward movement: Land with an active flat foot contact in the starting position, with hips and knees stiff, and immediately repeat.

Note: This drill should be performed with little horizontal (forward or backward) or lateral movement.

COMMON ERRORS

- **Flexing the hips and knees on landing to increase vertical force, which increases ground contact time and changes the nature of the ankle-dominant exercise**
- **Not dorsiflexing the ankle in flight; this makes athletes land on their toes, which changes the nature of the ground contact**



2 ANKLE HOP

Intensity level: Medium

Direction: Vertical

Starting position: Get into a comfortable, upright stance on one foot. The non-hopping leg is held in a stationary position with the knee flexed during the exercise.

Arm action: None or double arm

Preparatory movement: Begin with a slight countermovement.

Upward movement: Using the balancing foot, hop upward, with primary motion at the ankle joint.

Downward movement: Have the ankle dorsiflexed in flight to prepare for an active flat foot contact. (*Note:* The second photo does not yet show the left foot in the needed position for landing.) Land in the

starting position with the hips and knees stiff, and immediately repeat. Repeat with the opposite leg after a brief rest.

Note: This drill should be performed with little horizontal (forward or backward) or lateral movement.

COMMON ERRORS

- **Flexing the hips and knees on landing to increase vertical force; this increases ground contact time and changes the nature of the ankle-dominant exercise**
- **Landing off-balance so that rebound hops become multidirectional rather than vertical**



3

SQUAT JUMP

Intensity level: Low

Direction: Vertical

Starting position: Get into a squat position (thighs slightly above parallel with the ground) with feet hip- to shoulder-width apart. Interlock the fingers and place the hands behind the head.

Arm action: None

Preparatory movement: None

Upward movement: Explosively jump up to a maximum height. Extend the body as tall as possible.

Downward movement: Ankles should be dorsiflexed in flight and ready for flat foot landing. (*Note:* The second photo does not yet show the feet in the needed position for landing.) Land in the squat position, and immediately repeat the jump.

COMMON ERRORS

- **Landing and immediately jumping again before resetting to achieve squat depth for the subsequent jump**
- **Lack of hip and knee extension, meaning the body is not fully extended in flight**



4 JUMP AND REACH

Intensity level: Low

Direction: Vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm with reach at the top of the jump

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up and reach for an object or target.

Downward movement: Ankles should be dorsiflexed in flight and ready for an active flat-foot landing. Land in the starting position and immediately repeat the jump.

Note: Emphasis is on vertical height with minimal delay between jumps. Also, this drill should be performed with little horizontal

(forward or backward) or lateral movement.

COMMON ERRORS

- **Not coordinating arm action with the jump; everything should move upward at the same time**
- **Trunk not being vertical at takeoff, meaning that vertical force generation is compromised and forward movement will occur**



5 TUCK JUMP

Intensity level: Medium

Direction: Vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up. Pull the knees to the chest, quickly grasp the knees with both hands, and release before landing. Pull the ankles toward the knees in flight to prepare for an active flat-foot landing.

Downward movement: Land in the starting position, and immediately repeat the jump.

COMMON ERRORS

- **Pulling the knees up rather than pushing into the ground maximally then tucking knees**
- **Failure to perform a countermovement on subsequent repetitions; this decreases ground contact time but reduces vertical force generation**



6 SPLIT SQUAT JUMP

Intensity level: Low

Direction: Vertical

Starting position: Get into a lunge position with one leg forward (hip and knee joints flexed approximately 90 degrees) and the other behind the midline of the body with the trunk vertical.

Arm action: Double arm or none

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up, using the arms to assist as needed. Maximum height and power should be emphasized.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. (*Note:* The second photo does not yet show the feet positioned for landing.) When landing, maintain the lunge position (with the same leg forward) and immediately repeat the jump. After completing a set, rest and switch front legs.

COMMON ERRORS

- **Lack of coordinated triple extension (hips, knees, and ankle plantarflexion) in the takeoff, reducing the vertical force production**
- **Landing with the trunk leaning too far forward so the appropriate split squat position is not achieved**



7

CYCLED SPLIT SQUAT JUMP

Intensity level: Medium

Direction: Vertical

Starting position: Get into a lunge position with one leg forward (hip and knee joints flexed approximately 90 degrees) and the other behind the midline of the body.

Arm action: Double arm or none

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up, using the arms to assist as needed. While off the ground, switch the position of the legs.

Maximum height and power should be emphasized.

Downward movement: Dorsiflex the ankle of the lead landing leg in preparation for an active flat-foot landing. When landing, maintain the lunge position (opposite leg forward) and immediately repeat the jump. Be sure not to absorb ground reaction force and sink too deep into the lunge, because the SSC may not be able to effectively contribute to subsequent jumps.

COMMON ERRORS

- **Overemphasizing a quick split rather than vertical drive to create time for the leg switch to occur prior to landing**
- **Landing with the trunk leaning too far forward so the necessary split squat position is not achieved**



8

TUCK HOP

Intensity level: High

Direction: Vertical

Starting position: Get into a comfortable, upright stance on one foot. The non-hopping leg is held in a stationary position with the knee flexed during the exercise.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively hop up. Pull the knee of the hopping leg to the chest, grasp the knee with both hands, and release before landing.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. Land in the starting position, and immediately repeat the jump using the same leg. Complete a set, then repeat with the opposite leg after a brief rest.

COMMON ERRORS

- **Lack of vertical height achieved before achieving the tuck position**
- **Poorly coordinated sequencing of the arm action relative to the upward drive from the legs**



9 PIKE JUMP

Intensity level: Medium

Direction: Vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up. Keeping the legs straight and together, try to lift them to the front and try to touch the toes with the hands.

Downward movement: Land in the starting position, and immediately repeat the jump.

COMMON ERRORS

- **Overemphasis on bringing the legs forward rather than pushing into the ground to achieve vertical height**
- **Poor landing position due to lack of recovery from the pike**



Standing Jumps

10 VERTICAL JUMP

Intensity level: Low

Direction: Vertical

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump up, using both arms to assist, and reach for a target.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. (*Note:* The second photo does not yet show the feet in the ideal position for landing, because the athlete is still rising.) Land in the starting position, and immediately repeat the action for repetitions.

COMMON ERRORS

- **Poor sequencing of the forward and upward arm (shoulder) flexion in sequence with lower limb triple extension, which results in a coordination loss of upward momentum**
- **Inappropriately deep countermovement, which increases the amortization phase and therefore the ground contact time between jumps**



11

VERTICAL HOP

Intensity level: Medium

Direction: Vertical

Starting position: Get into a comfortable, upright stance on one foot. The non-jumping leg is held in a stationary position with the knee flexed.

Arm action: Double arm

Preparatory movement: Begin with a countermovement on the hopping leg. Bring the other leg behind the body.

Upward movement: Explosively hop up, driving the non-hopping leg forward and up to assist momentum. Use both arms to assist, and

reach for a target.

Downward movement: Dorsiflex the ankle in preparation for an active flat-foot landing. Land in the starting position, and immediately repeat the action for repetitions. Repeat with the opposite leg after a brief rest.

COMMON ERRORS

- **Not using the non-hopping leg in a coordinated manner to increase vertical momentum**
- **Not sequencing the arm action with the upward movement of the lower limb so vertical momentum is lost**

12 JUMP OVER BARRIER

Intensity level: Medium

Direction: Horizontal and vertical

Equipment: A barrier such as a cone or hurdle

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Take off with a full triple extension of hips, knees, and ankles. (*Note:* The second photo should show the athlete in a tall position to demonstrate full triple extension after takeoff. It also does not yet show the feet in the correct position for landing.) Jump over the barrier with both legs, maximizing the height of the center of mass and using only sufficient hip and knee flexion to clear the barrier. Keep the knees and feet together without lateral deviation.

Downward movement: Dorsiflex the ankles in preparation for a stable flat-foot landing. Land in the starting position and reset. Repeat until the set is complete.

Note: The height of the barrier should be progressively increased (e.g., from a cone to a hurdle to a higher hurdle) as the athlete adapts to the intensity of the drill.

COMMON ERRORS

- **Hurdle height insufficient to challenge the athlete to maximally push from the ground to achieve an extended position above the hurdle**
- **Inappropriate landing position whereby the athlete lands in an unstable position**



13 STANDING LONG JUMP

Intensity level: Low

Direction: Horizontal

Starting position: Get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Explosively jump forward and up, using both arms to assist, with a goal of achieving maximal horizontal distance.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. Land on both feet. Allow recovery time between jumps.

Note: This is typically a single ballistic effort, with several repetitions in a set. However, it can be done as multiple jumps in sequence. Care should be taken so that the forward momentum of the athlete does not cause overrotation in the sagittal plane. Coach the athlete to jump up and out; otherwise the tendency is for too much horizontal momentum for the athlete to control.

COMMON ERRORS

- **Inefficient takeoff angle, resulting in suboptimal flight distance; the lean should be far enough forward to optimize horizontal propulsion but not so far that the center of mass is too anteriorly located to optimize ground reaction forces**
- **Lack of full extension in the legs and full flexion in the shoulders at takeoff, which reduces the opportunity for ground reaction impulse to be created**



Multiple Hops and Jumps

14 MULTIPLE HORIZONTAL LOW HURDLE JUMP

Intensity level: Medium

Direction: Horizontal and vertical

Starting position: Get into a comfortable upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Jump as far forward as possible and over the first barrier. Coach the athlete to jump “upward and outward” and “get as tall as possible over the hurdle” rather than tuck jump to clear and not extend the body fully.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. Land in the starting position, and

immediately repeat the jump, optimizing ground contact time to control impulse production.

Note: A low hurdle height should encourage low flight time and lower ground contact time. The exercise can be increased in intensity with a higher hurdle height, but this will typically influence the nature of the ground contact.

COMMON ERRORS

- **Incorrect hurdle height or a less-than-maximal push from the ground to achieve full extension position when clearing the hurdle**
- **Inappropriate landing mechanics that increase ground contact time in between hurdles**



15 DOUBLE-LEG ZIGZAG JUMP

Intensity level: Medium

Direction: Diagonal

Equipment: Place 8 to 12 hurdles 18 to 24 inches (46-61 cm) apart in a zigzag pattern.

Starting position: Get into a comfortable, upright stance with feet hip-width apart. Stand on the outside of the first hurdle. Elbows should be flexed at 90 degrees and held at the sides of the body.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

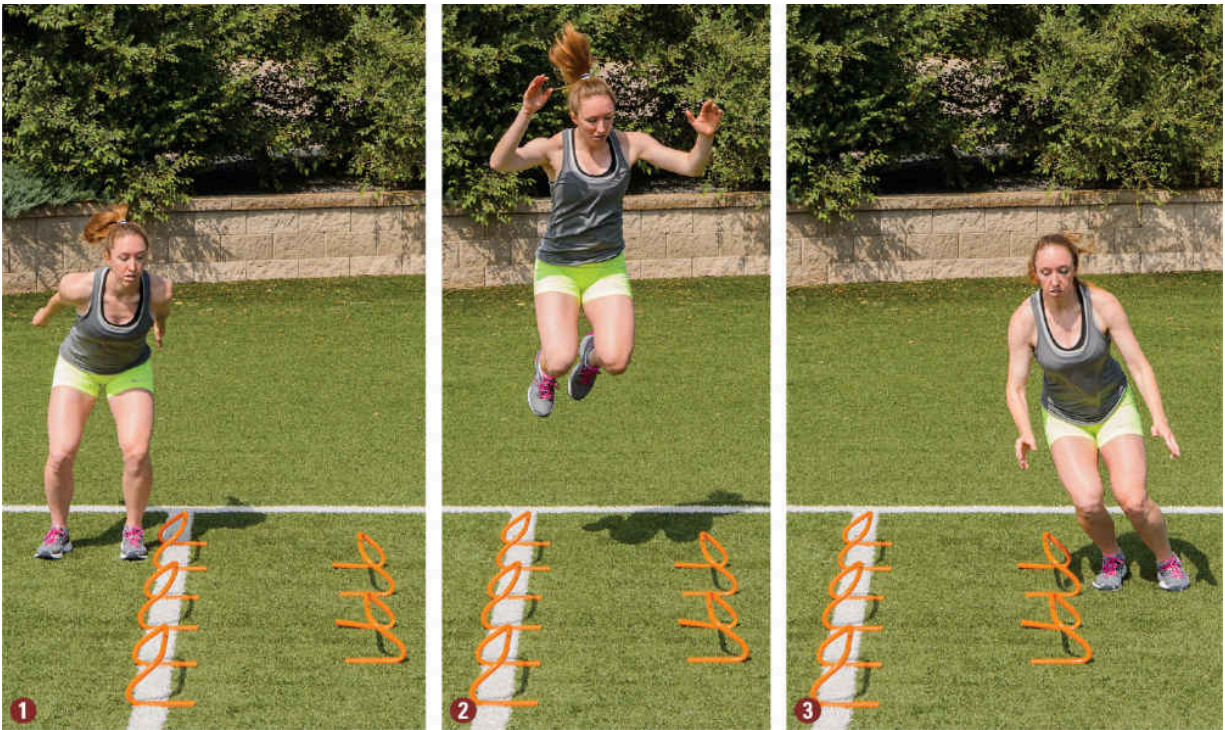
Upward movement: Jump from the outside of the first hurdle to the outside of the second hurdle, keeping the hips and shoulders facing the direction of travel.

Downward movement: In flight, dorsiflex the ankles in preparation for an active flat-foot landing. Immediately on landing on the outside of the second hurdle, change direction and jump diagonally over the second hurdle to the outside of the third hurdle. Continue jumping over all the hurdles.

Note: Hurdle height can significantly alter the intensity and in-flight body position of the athlete. The athlete should be as extended as possible to clear the hurdle. (*Note:* The second photo shows the athlete clearing the hurdle too high relative to the hurdle height and with excessive knee flexion.) Correct execution will result in the hips and knees being as stiff as possible on landing to optimize ground contact. Intensity of the drill can be significantly increased by performing hops with only one leg.

COMMON ERRORS

- **Not adjusting the body angle coming into a landing to keep the center of mass going in the desired direction of the next jump; this significantly increases ground contact time**
- **Not “attacking the ground” from a dorsiflexed ankle position, which increases ground contact time**



16 LINEAR HOP

Intensity level: Medium

Direction: Horizontal and vertical

Starting position: Get into a comfortable, upright stance on one foot. The non-jumping leg is held in a stationary position with the hip and knee flexed during the exercise.

Arm action: Double arm

Preparatory movement: Begin with a countermovement on the hopping leg. Bring the other leg behind the body.

Upward movement: Explosively hop forward, driving from the hip, knee, and ankle, using both arms and the opposite knee to assist the movement.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing. (*Note:* The second photo does not yet show the feet in the needed position for landing.) Land in the starting position, use an active flat-foot contact to optimize the ground contact time, and immediately repeat the hop for the desired

number of repetitions. Repeat with the opposite leg after a brief rest.

COMMON ERRORS

- **Excessive flexion of the hip, knee, and ankle on landing, which unnecessarily increases the amortization phase and ground contact time**
- **Hopping with the weight too far forward in the foot, which reduces ground reaction forces and limits optimal force production to maximize flight distance**



17

MULTIPLE BARRIER HOP

Intensity level: Medium to high

Direction: Horizontal and vertical

Equipment: Two to 10 barriers such as cones or hurdles

Starting position: Facing the first barrier, get into a comfortable, single-leg upright stance with the contralateral hip and knee flexed to 90 degrees.

Arm action: Double arm

Preparatory movement: Begin with a countermovement on the hopping leg. Bring the other leg behind the body.

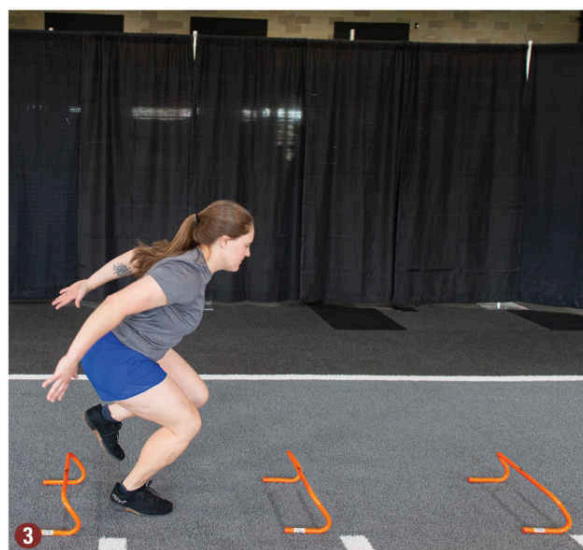
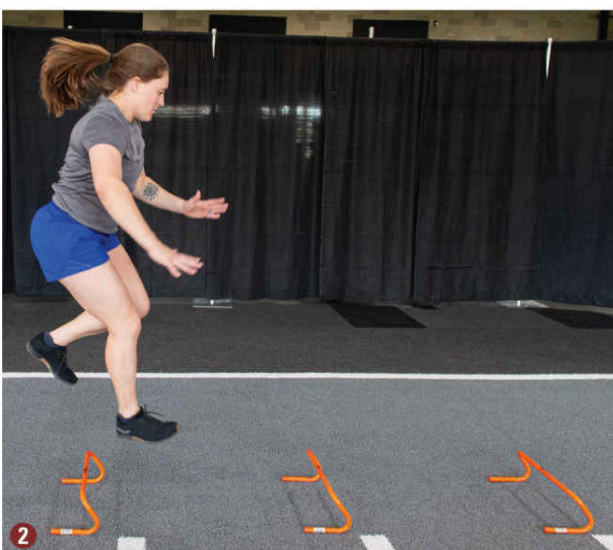
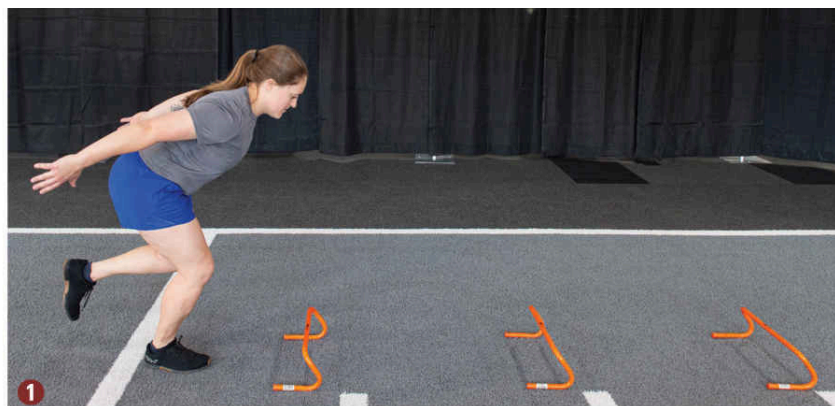
Upward movement: Explosively hop forward, driving from the hip, knee, and ankle over the first barrier as tall as possible, using necessary hip and knee flexion to clear the barrier.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing in the starting position, and immediately repeat the jump over the second barrier.

Note: Intensity level of this exercise can be increased from medium to high by progressively increasing the height of the barrier (e.g., from a cone to a hurdle). Increasing the intensity of the hop may increase the ground contact time and the amortization phase.

COMMON ERRORS

- **Incorrect hurdle height or a less-than-maximal push from the ground to achieve full extension position when clearing the hurdle**
- **Inappropriate landing mechanics, which increases ground contact time between hurdles**



18 LATERAL BARRIER JUMP

Intensity level: Medium

Direction: Lateral and vertical

Equipment: A barrier such as a cone or hurdle

Starting position: With the barrier to one side, get into a comfortable, upright stance with feet hip-width apart.

Arm action: Double arm

Preparatory movement: Begin with a countermovement.

Upward movement: Leaning out of the countermovement in the direction of travel, use a powerful triple extension to jump over the barrier as tall as possible with both legs, using necessary hip and knee flexion to clear the barrier. (*Note:* The second photo shows the